

GEODE: Geometry-Guided Discrete Diffusion for Open-Vocabulary 3D Scene Graph Generation

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Abstract

*3D Scene Graph Generation (3DSGG) seeks to build a structured representation of a 3D scene by localizing objects and predicting their pairwise relations. Recent efforts push beyond the closed-set to open-vocabulary (OV) recognition by leveraging large vision-language models (VLMs). However, these pipelines inherit VLMs' weakness in spatial reasoning, yielding brittle predicate predictions under near-depth ambiguities and symmetric layouts. We introduce **GEODE**, a **GEOM**etry-guided **Discrete** diffusion framework that jointly denoises node geometry and edge predicates over tokenized scene graphs. Our key idea is to generate a globally geometry-constrained scene graph with a discrete graph diffusion process. Concretely, we vector-quantize per-object geometric attributes (center/size/orientation) and predicate semantics into compact codebooks, and perform categorical diffusion with an intra-step geometry to relation schedule: at each reverse step, geometry tokens are sampled first and then condition predicate updates, enforcing geometric consistency throughout generation. A Geometry-aware Graph Transformer tightly couples node-edge reasoning via relation-conditioned self-attention and an orientation-aware decomposition of edge updates, enabling precise spatial grounding without sacrificing semantic expressivity. On 3DSSG, **GEODE** sets a new state of the art in both closed-set and OV regimes, with particularly large gains on spatial-only evaluation under a strict top-1 criterion, e.g., with 19 point improvements over the SOTA RelationField.*

1. Introduction

3D Scene Graph Generation (3DSGG)[5, 28, 42, 57–60] builds a graph where nodes are detected objects and edges

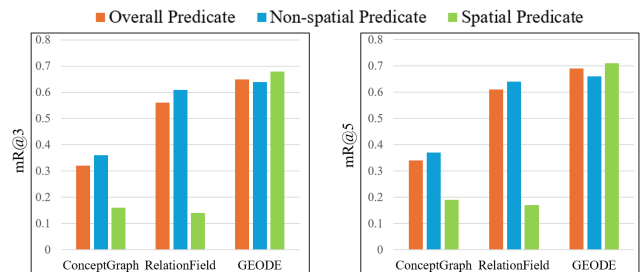


Figure 1. Comparison of our proposed **GEODE** and two representative SOTA VLM-based open-vocabulary 3DSGG models [15, 28] on spatial predicate classification task on 3DSSG [56].

encode semantic relations (spatial, comparative, functional). This structured representation turns raw geometric/visual cues into a machine-interpretable representation that benefits embodied navigation [49], VQA [68], image retrieval [50], and manipulation [47].

Earlier 3DSGG systems are largely *closed-set*, fixing object and predicate vocabularies and thus limiting transfer across datasets and domains [11, 20, 26, 58]. Recent open-vocabulary (OV) approaches [15, 28, 29] inject VLM priors to relax the label space, but VLMs are notoriously weak on spatial relations [9, 24, 25]. To quantify this gap, we run a spatial-centric diagnostic (Sec. 5.2) that splits predicates into spatial vs. non-spatial and reports mR@k. As shown in Fig. 1, VLM-based baselines (e.g., ConceptGraph [15], RelationField [29]) retain moderate recall on non-spatial predicates but collapse on the spatial subset, e.g., spatial predicate in terms of mR@3/5 is typically 2–4× lower than their non-spatial counterpart, with errors concentrated in near-depth ambiguities.

Why does this happen? We hypothesize two primary causes: (i) **Data deficiency**. VLMs [31, 43] are trained on web-scale image-text pairs that rarely supply precise or systematic 3D geometric descriptions, limiting explicit 3D alignment capabilities. (ii) **Inference paradigm**. Most

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pipelines infer spatial relations predominantly through vision–text alignment, encouraging pattern matching instead of explainable reasoning over explicit 3D attributes (coordinates, extents, orientations, contact/containment, etc.). Naïvely injecting 3D cues into visual features often yields shortcut learning [13], where models memorize dataset-specific coordinate/co-occurrence statistics and fail under geometric shifts [59].

We posit that spatial predicates should be *constrained by explicit geometry*, not inferred solely from vision–text alignment. If A is “left of” B , their coordinates must satisfy the constraint (i.e., the x-coordinate of A should be smaller than that of B); relation decisions should be validated against such constraints throughout generation. This motivates our core research question: *Can we compel models to infer spatial relations from geometric attributes rather than remembering visual–text alignment statistics?*

To answer this question, we reformulate 3DSGG as *geometry-guided* discrete generation over tokenized scene graphs. Nodes carry explicit geometric attributes; edges are predicates. Starting from masked/noisy tokens, a progressive denoiser recovers geometry and relations in tandem, and within each step conditions predicates on the freshly recovered geometry, thereby encoding geometric consistency along the trajectory and suppressing shortcuts.

Building on these insights, we present **GEODE** (GEOMETRY-guided DISCRETE diffusion) for open-vocabulary 3DSGG. We first quantize per-node geometry (center/size/orientation) and predicate semantics into discrete vocabularies using lightweight VQ-VAE tokenizers [7, 33, 45]; detector-provided classes remain fixed and provide node semantics throughout generation (Sec. 4.1). Similar to diffusion-based image generation [24, 46, 51], we begin by applying a forward corruption process to add noise and mask the discrete tokens using a noising kernel matrix [46] (Sec. 4.2). A Geometry-aware Graph Transformer (G^2T) denoiser operates entirely in the discrete space (Sec. 4.3): node updates inject pairwise cues via edge-conditioned attention (FiLM) [2, 3, 33], and directed edge updates decompose modulated scores into interaction and orientation components with Mixture-of-Experts(MoE). Crucially, an *intra-step* geometry→relation schedule first samples geometry tokens and then predicts predicates conditioned on the newly recovered geometry, enforcing geometric consistency at every reverse step.

In summary, our contributions are fourfold:

- **New 3DSGG paradigm.** We cast scene graph construction as *geometry-conditioned discrete graph generation* over tokenized variables.
- **GEODE framework.** We introduce **GEODE** (GEOMETRY-guided DISCRETE diffusion) for 3DSGG, a discrete diffusion jointly denoising geometry and relations.
- **Geometry-aware denoiser.** We design a geometry-aware

graph transformer with relation-conditioned attention and orientation-aware edge updates with MoE.

- **SOTA results.** On standard 3DSGG benchmarks, GEODE surpasses state-of-the-art methods in both closed-set and open-vocabulary settings, with particularly large improvements on spatial predicates.

2. Related Work

Open-Vocabulary 3D Scene Graph Generation. 3D scene graphs aim to recognize objects and model semantic relations to form a structured graph of a scene. Early methods follow a closed-vocabulary paradigm, requiring extensive labeled data and failing to generalize beyond predefined classes [17, 19, 21, 60]. Recent works leverage multimodal/vision–language models to move toward open-vocabulary 3D scene graphs [22, 28, 70]. However, these approaches often under-address VLMs’ limitations in spatial reasoning, leading to unstable predictions for spatial relations (e.g., layout/support) [9, 24]. To remedy this, we (to our knowledge, for the first time) introduce a discrete diffusion framework for open-vocabulary 3D scene graph generation. We propose a node–attribute–edge triad with independent masking and collaborative denoising to jointly model geometric attributes and relation semantics, and we explicitly encode spatial geometry in node attributes to strengthen spatial reasoning.

Discrete Diffusion Models. Diffusion models have achieved strong results in continuous domains [10, 24, 30, 51, 66] and have been extended to discrete state spaces: D3PM generalizes diffusion via structured transition matrices [1]. For graphs, DiGress introduces discrete graph diffusion that treats noise as independent graph edits and couples independent masking with collaborative denoising across attributes [55]. Subsequent work advances structural constraints and latent formulations: ConStruct enforces hard constraints throughout generation [38], and Latent Graph Diffusion unifies generation and prediction across levels and tasks [71]. Scene graph generation is inherently iterative [64], and nodes/edges are categorical variables [34, 53], making it a natural fit for discrete diffusion. Building on these insights, we adopt discrete diffusion for open-vocabulary 3D scene graphs, combining the proposed triad and quantization design to achieve geometry–relation coupling and open-vocabulary generalization.

3. Preliminaries

Problem Statements. We cast 3D scene graph generation (3DSGG) as *conditional discrete graph generation*. Given a scene descriptor x (e.g., multi-view features or a fused point cloud) and detected objects $\mathcal{V} = \{v_i\}_{i=1}^N$, the goal is to sample a typed, directed multigraph $G = (\mathcal{V}, \mathcal{E}, \mathcal{G})$, where $\mathcal{G} = \{g_i\}$ are explicit 3D attributes (center/size/location,

etc.) and $\mathcal{E}$ contains possibly multiple edge types per ordered pair. We discretize node geometry and edge predicates into token sequences, $Z_G \in [K_G]^{N \times T_G}$ and $Z_E \in [K_E]^{|\mathcal{P}| \times T_E}$, with $\mathcal{P} = \{(i, j, c) \mid i \neq j, c \in \{1, \dots, C\}\}$. The generator defines a reverse Markov chain over graph-token states $S_t = (Z_{G,t}, Z_{E,t})$ that denoises from an all-[MASK] prior $q(S_T)$ to a clean graph S_0 :

$$p_\theta(G \mid \mathcal{V}, x) = \sum_{S_{1:T-1}} \left[\prod_{t=1}^T p_\theta(S_{t-1} \mid S_t, \mathcal{V}, x) \right] q(S_T), \quad (1)$$

where the time step t is from 0 to T .

Discrete Diffusion for Categorical Variables. For a categorical variable z with K classes (e.g., a node or edge label), we follow D3PM [1] and define a discrete forward corruption using a *categorical* Markov chain. Represent each state as a one-hot row vector. We augment the alphabet with a single absorbing token [MASK] so the total size is $K' = K+1$. The forward transition at step t is

$$q(z_t \mid z_{t-1}) = \text{Cat}(z_t; p = z_{t-1} Q_t), \quad (2)$$

where $Q_t \in \mathbb{R}^{K' \times K'}$ is row-stochastic. Using per-step coefficients α_t (stay), β_t (uniform jump within the K original classes), and γ_t (mask), a convenient instance is

$$Q_t = \begin{bmatrix} \alpha_t + \beta_t & \beta_t & \cdots & \beta_t & \gamma_t \\ \beta_t & \alpha_t + \beta_t & \cdots & \beta_t & \gamma_t \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \beta_t & \beta_t & \cdots & \alpha_t + \beta_t & \gamma_t \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}, \quad (3)$$

and $\alpha_t + (K-1)\beta_t + \gamma_t = 1$, $\alpha_t, \beta_t, \gamma_t \geq 0$. so a token is (i) kept with prob. α_t , (ii) replaced uniformly among the other $K-1$ classes with total prob. $(K-1)\beta_t$, or (iii) sent to [MASK] with prob. γ_t . Accumulating noise yields the closed-form marginal

$$q(z_t \mid z_0) = \text{Cat}(z_t; p = z_0 \bar{Q}_t), \quad \bar{Q}_t = Q_1 Q_2 \cdots Q_t. \quad (4)$$

The exact forward posterior used in training is

$$q(z_{t-1} \mid z_t, z_0) = \text{Cat}\left(z_{t-1}; p = \frac{(z_t Q_t^\top) \odot (z_0 \bar{Q}_{t-1})}{z_0 \bar{Q}_t z_t^\top}\right), \quad (5)$$

with $\odot$ denoting element-wise multiplication. A neural model $p_\theta(z_{t-1} \mid z_t)$ is trained to approximate (5) (typically via cross-entropy over categorical logits). The masking schedule $\{\gamma_t\}$ gradually drives states toward [MASK], enabling robust denoising from partially observed graphs.

4. Method

We present **GEODE**, a geometry-conditioned, multi-relation 3D scene-graph generation framework built on a

unified discrete-diffusion backbone. Our model unifies object nodes, explicit geometry and multi-type relations in a single probabilistic formulation and enforces **geometric consistency** by predicting relations conditioned on recovered geometry. Practically, both geometry and relation semantics are tokenized with VQ models and denoised with a categorical diffusion process (Sec. 3). Below we describe scene graph discretization, specify the discrete diffusion generative model, detail the denoiser architecture, and give the training and inference procedures, all using the notation introduced in Sec. 3.

4.1. Scene Graph Quantization

We formulate scene-graph construction as conditional generation over discrete tokens. For *nodes*, we quantize object geometry (center, size, orientation), and for *edges*, we quantize predicate semantics. Object classes are provided by an off-the-shelf detector [28, 41, 44, 61] and serve as fixed semantics, where node embeddings condition on class labels without resampling during generation. Below, we describe the node and edge tokenization processes.

Node geometry tokenization. Each node’s geometry is decomposed into three attribute groups: $\mathbf{g}_i = (g_{i,\text{pos}}, g_{i,\text{size}}, g_{i,\text{ang}})$, and for each attribute $w \in \{\text{pos}, \text{size}, \text{ang}\}$, we assign a dedicated codebook $\mathcal{C}_w$. The attribute $g_{i,w}$ is encoded to a latent vector $\mathbf{z}_{i,w}$, which is quantized via nearest-neighbor lookup:

$$k_{i,w} = \arg \min_{k \in [K_w]} \|\mathbf{z}_{i,w} - \mathbf{e}_{w,k}\|_2^2, \quad \mathbf{q}_{i,w} = \mathbf{e}_{w,k_{i,w}},$$

where $\mathcal{C}_w = \{\mathbf{e}_{w,k}\}_{k=1}^{K_w}$. The local indices are mapped to a global geometry vocabulary with disjoint offsets O_w :

$$Z_G[i, w] = k_{i,w} + O_w \in [K_G], \quad K_G = \sum_w K_w.$$

Training uses the standard VQ-VAE objective [54].

Edge relation tokenization. For each ordered triple $p = (i, j, c) \in \mathcal{P}$ (subject v_i , object v_j , and candidate predicate c), we form a natural language template (e.g., y_i is c y_j), where y_i and y_j are object labels from the detector. The text is encoded using a frozen CLIP text encoder $\mathcal{T}$ to obtain a 512-dimensional embedding: $\mathbf{e}_{\text{clip}}^{(p)} = \mathcal{T}(\text{template}(y_i, c, y_j))$, which is adapted via a lightweight residual adapter $\mathcal{A}$: $\hat{\mathbf{e}}^{(p)} = \mathcal{A}(\mathbf{e}_{\text{clip}}^{(p)}) = \mathbf{e}_{\text{clip}}^{(p)} + \alpha \text{MLP}\left(\text{LayerNorm}(\mathbf{e}_{\text{clip}}^{(p)})\right)$, where α is a learnable temperature parameter. An encoder ϕ_E produces the latent $\mathbf{z}^{(p)} = \phi_E(\hat{\mathbf{e}}^{(p)})$, which is quantized using a relation codebook $\mathcal{C}_{\text{rel}} = \{\mathbf{e}_k\}_{k=1}^{K_E}$:

$$k^{(p)} = \arg \min_{k \in [K_E]} \|\mathbf{z}^{(p)} - \mathbf{e}_k\|_2^2, \quad \mathbf{q}^{(p)} = \mathbf{e}_{k^{(p)}}.$$

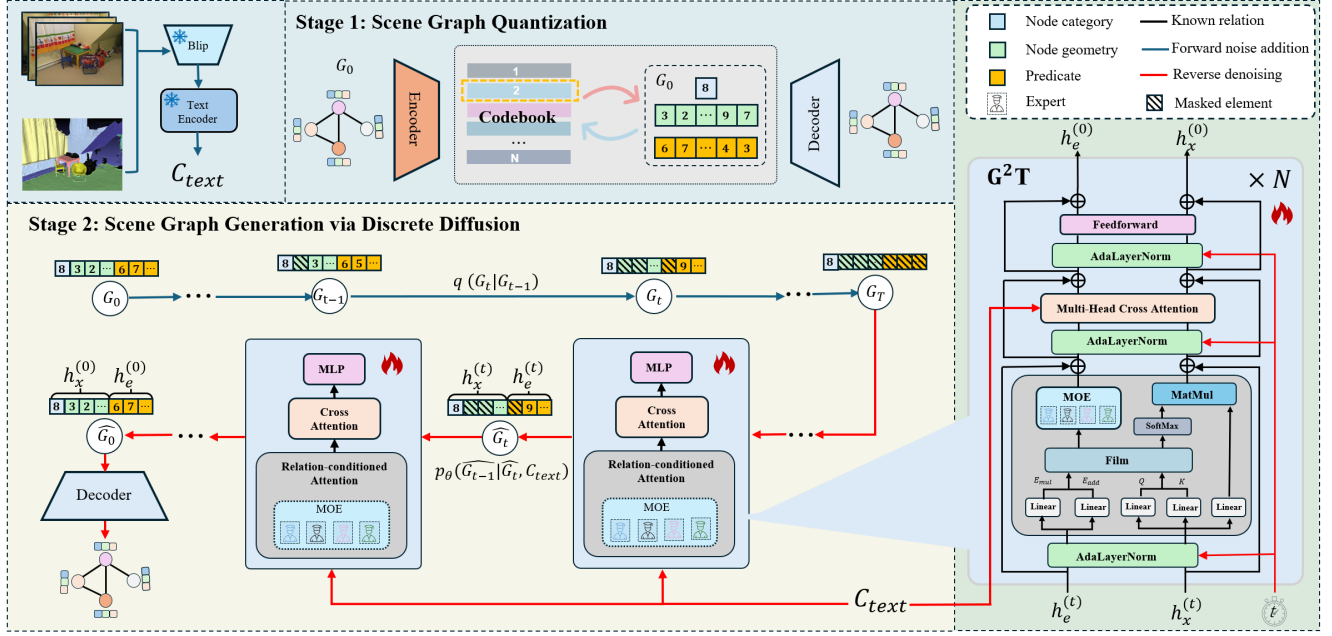


Figure 2. **Overview of GEODE.** GEODE consists of two stages: (i) a VQ-VAE discretizes the 3D scene graph into node-geometry and edge-predicate tokens, and BLIP encodes a global scene description; (ii) a discrete diffusion process over these tokens uses a geometry-aware G^2T denoiser, conditioned on geometry and text, to iteratively recover relation tokens and enforce geometry–relation consistency.

The edge token is given by: $Z_E[p] = k^{(p)} \in [K_E]$. The edge tokenizer is trained with the VQ-VAE loss [18, 33, 45].

Image condition extraction. To condition the denoiser on a global scene representation, we first extract multi-scale 2D/3D features from the paired image-text inputs. These features are processed by a captioning model [31, 32] to generate a comprehensive scene description. The caption is encoded by the CLIP text encoder, and a lightweight MLP adapter is used to produce a global condition embedding C_{text} , which is fed to the denoiser.

4.2. Scene Graph Generation via Discrete Diffusion

We generate 3D scene graphs with categorical diffusion over unified geometry–relation tokens: a time-dependent kernel Q_t corrupts tokens, and a graph transformer predicts clean-token posteriors that, reweighted by Q_t , drive sampling from $T \rightarrow 0$. Each reverse step updates geometry then relations under structural masks/clamping, enforcing geometric consistency.

4.2.1. Forward Corruption Processing

At each timestep t , tokens are corrupted independently with a time-dependent categorical kernel Q_t (Eq. (3)). The joint forward distribution factorizes across modalities:

$$q(Z_t | Z_0) = q(Z_{G,t} | Z_{G,0}) q(Z_{E,t} | Z_{E,0}), \quad (6)$$

and within each modality across token positions. Concretely, Q_t mixes (i) a *stay* event with probability α_t , (ii) a

uniform jump over the vocabulary with probability β_t , and (iii) replacement by a special [MASK] token with probability γ_t . The [MASK] symbol serves as a learned placeholder for missing content, enabling structured inpainting during the reverse pass. The forward algorithm is presented in the upper block of Alg. 1.

4.2.2. Reverse Denoising Processing

At each diffusion step t , we denoise the joint token state $S_t = (Z_{G,t}, Z_{E,t})$ with a graph transformer that predicts *clean* (time-0) token distributions. These predictions are combined with the forward corruption kernel Q_t to form an exact one-step posterior over z_{t-1} . To enforce geometric consistency, every step proceeds in two sub-steps—first updating geometry, then relations conditioned on the newly sampled geometry—while respecting structural constraints via masked candidate sets and clamping any known tokens.

Reverse Parameterization. We instantiate a graph transformer f_θ that consumes the corrupted tokens S_t , detected nodes $\mathcal{V}$, the scene descriptor $\mathbf{x}$, and the timestep t . It outputs per-position logits over the clean vocabulary:

$$\pi_t^{(0)}(\cdot) = p_\theta(Z_0 = \cdot | S_t, \mathcal{V}, \mathbf{x}, t). \quad (7)$$

Given the forward kernel Q_t (Sec. 3), the one-step posterior at a token position is

$$p_\theta(z_{t-1} = a | z_t, S_t, \mathcal{V}, \mathbf{x}, t) = \frac{Q_t(z_t | a) \pi_t^{(0)}(a)}{\sum_{b \in [K]} Q_t(z_t | b) \pi_t^{(0)}(b)} \quad (8)$$

Algorithm 1 Discrete Diffusion for 3DSGG

Forward Process: Corruption of Graphs

- 1: **Input:** Scene descriptor $\mathbf{x}$, node set $\mathcal{V} = \{v_i\}_{i=1}^N$, clean graph $G = (\mathcal{V}, \mathcal{E}, \mathcal{G})$
- 2: **Output:** Noisy graph $G_T = (\mathcal{V}, \mathcal{E}_T, \mathcal{G}_T)$
- 3: **for** $t = 0$ to T **do**
- 4: Sample geometry tokens: $Z_{G,t} \sim \text{Cat}(Q_{G,t})$
- 5: Sample relation tokens: $Z_{\mathcal{E},t} \sim \text{Cat}(Q_{\mathcal{E},t})$
- 6: Apply corruption by masking: $Z_{G,t}, Z_{\mathcal{E},t} \rightarrow [\text{MASK}]$
- 7: **end for**
- 8: **Return:** $G_T = (\mathcal{V}, \mathcal{E}_T, \mathcal{G}_T)$

Reverse Process: Denoising the Graphs

- 1: **Input:** Noisy graph $G_T = (\mathcal{V}, \mathcal{E}_T, \mathcal{G}_T)$, scene descriptor $\mathbf{x}$
 - 2: **Output:** Clean graph $G_0 = (\mathcal{V}, \mathcal{E}_0, \mathcal{G}_0)$
 - 3: **for** $t = T$ to 1 **do**
 - 4: Predict clean distribution: $\pi_t^{(0)} = p_\theta(Z_0 \mid Z_t, \mathcal{V}, \mathbf{x}, t)$ by Eq. (7)
 - 5: Compute posterior $p_\theta: Q_t(z_t \mid a) \cdot \pi_t^{(0)}(a)$ by Eq.(8)
 - 6: Sample $Z_{t-1} \sim \text{Gumbel}(\log p_\theta(z_{t-1} \mid z_t))$
 - 7: **end for**
 - 8: **Decode:** $Z_{G,0}, Z_{\mathcal{E},0} \rightarrow \hat{\mathcal{G}}, \hat{\mathcal{E}}$
 - 9: **Return:** $G_0 = (\mathcal{V}, \mathcal{E}_0, \mathcal{G}_0)$
-

where $a \in [K]$ indexes the position's vocabulary.

Intra-step Geometry→Relation Updating. To propagate geometric constraints into relation decisions within the same step, we factor each reverse update into two sub-steps:

- (i) $Z_{G,t-1} \sim p_\theta(Z_{G,t-1} \mid S_t, \mathcal{V}, \mathbf{x}, t)$
- (ii) $Z_{\mathcal{E},t-1} \sim p_\theta(Z_{\mathcal{E},t-1} \mid Z_{G,t-1}, S_t, \mathcal{V}, \mathbf{x}, t)$,

Relation logits are produced after sampling $Z_{G,t-1}$ so that predicates explicitly condition on the freshly recovered geometry. During the reverse step, we apply the Gumbel function [4, 52] for sampling. The reverse algorithm is presented in the bottom block of Alg. 1.

4.3. Architecture of the Denoiser

We build upon discrete graph generation [6, 55] and introduce the *Geometry-aware Graph Transformer* ($\mathbf{G}^2\mathbf{T}$), which jointly reasons over object geometry (nodes) and relations (edges) while explicitly coupling their updates. At each diffusion step t , inputs are split into *conditions* and a *noisy state*. Conditions comprise clean node classes $\mathbf{x}_0$ (from a detector) and an optional scene description $\mathbf{C}_{\text{text}}$, injected via cross-attention as in text-to-image generation [35, 40]. The noisy state comprises discretized node geometry $\mathbf{o}_t$ and relations $\mathbf{e}_t$. We decode discrete codes into continuous features:

$$\mathbf{h}_x^{(t)} = \text{Emb}_c(\mathbf{x}_0) \oplus \text{Emb}_o(\mathbf{o}_t), \quad \mathbf{h}_e^{(t)} = \text{Emb}_e(\mathbf{e}_t),$$

and apply time-adaptive normalization (AdaLN) [39, 63] with a time code $\mathbf{t}_{\text{emb}} = \phi(t)$:

$$\tilde{\mathbf{h}}_x = \text{AdaLN}(\mathbf{h}_x^{(\ell-1)}, \mathbf{t}_{\text{emb}}), \quad \tilde{\mathbf{h}}_e = \text{AdaLN}(\mathbf{h}_e^{(\ell-1)}, \mathbf{t}_{\text{emb}}).$$

Node updating with relation-conditioned attention. To bias attention toward geometrically/semantically plausible pairs, we modulate node self-attention with edge cues (FiLM-style [16]). From $\tilde{\mathbf{h}}_x$ we form queries, keys, values: $\mathbf{Q} = W_Q \tilde{\mathbf{h}}_x$, $\mathbf{K} = W_K \tilde{\mathbf{h}}_x$, $\mathbf{V} = W_V \tilde{\mathbf{h}}_x$, and from $\tilde{\mathbf{h}}_e$ we derive additive FiLM factors: $\mathbf{E}^{\text{mul}} = W_{e,\text{mul}} \tilde{\mathbf{h}}_e$, $\mathbf{E}^{\text{add}} = W_{e,\text{add}} \tilde{\mathbf{h}}_e$. Let $S_{ij} = \frac{\mathbf{Q}_i \mathbf{K}_j^\top}{\sqrt{d}}$ denote two nodes' similarity. The relation-modulated attention are

$$\hat{S}_{ij} = (1 + \mathbf{E}_{ij}^{\text{mul}}) S_{ij} + \mathbf{E}_{ij}^{\text{add}}, \quad \alpha_{ij} = \text{Softmax}_j(\hat{S}_{ij}),$$

and node states update by weighted aggregation:

$$\mathbf{h}_{x,i}^{(\ell)} = \mathbf{h}_{x,i}^{(\ell-1)} + W_o \left(\sum_{j=1}^N \alpha_{ij} \mathbf{V}_j \right). \quad (10)$$

Edge updating with MoE. Directed predicates require separating *interaction strength* from *orientation*. We decompose the modulated score matrix $\hat{\mathbf{S}} \in \mathbb{R}^{N \times N}$ into symmetric and antisymmetric parts:

$$\mathbf{S}_{\text{sym}} = \frac{1}{2}(\hat{\mathbf{S}} + \hat{\mathbf{S}}^\top), \quad \mathbf{S}_{\text{asym}} = \frac{1}{2}(\hat{\mathbf{S}} - \hat{\mathbf{S}}^\top). \quad (11)$$

To specialize updates for heterogeneous relation families (i.e., diverse types of predicates) while keeping computation tractable, we introduce a *Mixture-of-Experts* (MoE) [8, 12, 48] head on edges. The router leverages geometry, time and nodes' prior. For each ordered pair (i, j) , define the routing feature:

$$\mathbf{r}_{ij} = [\mathbf{h}_{e,ij}^{(\ell-1)} \oplus \rho_{ij} \oplus \mathbf{t}_{\text{emb}} \oplus u_{ij}], \quad (12)$$

where $\rho_{ij} = [\text{Emb}_o(\mathbf{o}_i) \oplus \text{Emb}_o(\mathbf{o}_j)]$ encodes geometry and $u_{ij} = [\text{Emb}_c(\mathbf{x}_0^i) \oplus \text{Emb}_c(\mathbf{x}_0^j)]$ is semantic priors. With E experts, the gate and sparse routing are $\mathbf{g}_{ij} = \text{softmax}(W_g \mathbf{r}_{ij}) \in \mathbb{R}^E$, $\mathcal{K}_{ij} = \text{TopK}(\mathbf{g}_{ij}, k)$. Thus, the expert selection score is $\alpha_{ij}^{(e)} = \frac{\exp(g_{ij}^{(e)})}{\sum_{e' \in \mathcal{K}_{ij}} \exp(g_{ij}^{(e')})}$. Each expert e produces edge updates that *respect* the decomposition in Eq. (11). Concretely,

$$(\Delta \mathbf{h}_{e,ij}^{(e)}) = \underbrace{W_{e,\text{sym}}^{(e)} [\mathbf{S}_{\text{sym},ij}]}_{\text{interaction}} + \underbrace{W_{e,\text{dir}}^{(e)} [\mathbf{S}_{\text{asym},ij}]}_{\text{orientation}},$$

where $W_{e,\text{sym}}^{(e)}$ and $W_{e,\text{dir}}^{(e)}$ are learnable per-expert linear maps that map scalar scores to edge-feature increments, optionally conditioned on $\mathbf{h}_{x,i}^{(\ell)}, \mathbf{h}_{x,j}^{(\ell)}$. The MoE-routed edge

update is then

$$\mathbf{h}_{e,ij}^{(\ell)} = \mathbf{h}_{e,ij}^{(\ell-1)} + \sum_{e \in \mathcal{K}_{ij}} \alpha_{ij}^{(e)} \Delta \mathbf{h}_{e,ij}^{(e)}.$$

Training Objectives. We train the denoiser f_θ by optimizing an objective function that combines the variational bound [39] from the diffusion process and an auxiliary node class reconstruction loss. For each token $u \in Z_G \cup Z_E$ at time step t , the loss $\mathcal{L}_t(u)$ is defined as:

$$\mathcal{L}_t(u) = \begin{cases} \text{KL}(q(u_{t-1} | u_t, u_0) \parallel p_\theta(u_{t-1} | \mathcal{C}_t)), & t > 1, \\ -\log p_\theta(u_0 | \mathcal{C}_1), & t = 1. \end{cases}$$

where $\mathcal{C}_t = (Z_t, \mathcal{V}, \mathbf{x}, t)$ is the set of conditions at step t . For $t > 1$, the loss minimizes the KL divergence between the forward posterior q and the model’s posterior p_θ . At the final step ($t = 1$), the loss is a reconstruction loss equivalent to the negative log-likelihood of the true token u_0 .

The variational bounds for geometry vb_o and relations vb_e are then computed as the expectations of $\mathcal{L}_t(u)$ over tokens in the geometry and relation spaces, respectively:

$$\text{vb}_o = \mathbb{E}_{\substack{t \sim \mathcal{U}[T], \\ u_0 \sim p_{\text{data}}, u_t \sim q(\cdot | u_0, t)}} \left[\frac{1}{|Z_G|} \sum_{u \in Z_G} \mathcal{L}_t(u) \right],$$

$$\text{vb}_e = \mathbb{E}_{\substack{t \sim \mathcal{U}[T], \\ u_0 \sim p_{\text{data}}, u_t \sim q(\cdot | u_0, t)}} \left[\frac{1}{|Z_E|} \sum_{u \in Z_E} \mathcal{L}_t(u) \right].$$

To prevent representation drift in node classes, we introduce an auxiliary negative log-likelihood (NLL) loss:

$$\mathcal{L}_{\text{aux-c}} = \mathbb{E}_{t, Z_t, \mathcal{V}, \mathbf{x}} [-\log p_\theta(\hat{c}_0 = c_0 | Z_t, \mathcal{V}, \mathbf{x}, t)].$$

The final training objective is the sum of these terms:

$$\mathcal{L} = \lambda_{\text{geo}} \cdot \text{vb}_o + \lambda_{\text{rel}} \cdot \text{vb}_e + \lambda_{\text{aux-c}} \cdot \mathcal{L}_{\text{aux-c}}.$$

where λ_{geo} , λ_{rel} and $\lambda_{\text{aux-c}}$ are hyperparameters that balance the contributions of each term.

5. Experiments

5.1. Experimental Setup and Datasets

Datasets. All experiments are conducted on 3DSSG [57], currently the only publicly available benchmark with large-scale semantic 3D scene graph annotations. For each reconstructed scene, 3DSSG provides complete graph-structured labels over object instances, semantic categories, and pairwise relations, and is widely adopted for 3D scene graph generation. The dataset contains roughly 1.5k indoor scenes, which are partitioned into about 4.1k sub-scenes following the official split (around 3.6k for training and the

rest for validation/testing), covering 160 object categories and 27 predicate types.

Evaluation protocol. We evaluate under both closed-set and open-vocabulary (OV) settings, with a focus on spatial reasoning. Following standard practice, we report Recall@K (R@K) and mean Recall@K (mR@K) for two tasks: (i) *Predicate Classification*, where ground-truth object boxes and labels are provided and the model predicts only predicates; and (ii) *Relationship Triplets*, where the model jointly detects objects and predicts predicates. To isolate geometric competence, spatial predicates are further evaluated with a strict top-1 criterion. In the OV setup, we additionally partition predicates into *spatial* and *non-spatial* subsets and report R@K/mR@K restricted to the spatial subset.

Baselines. In the *closed-set* setting, we compare against several recent state-of-the-art 3DSSG methods. In the *open-vocabulary* setting, we include representative OV approaches such as RelField [29] (CVPR ’25), CCL-3DSSG [5] (CVPR ’24), OW3DSSG [67] (AAAI ’26), ConGraph [15] (ICRA ’24), and Open3DSG [28] (CVPR ’24), etc. For OV evaluation with an emphasis on spatial predicates, we primarily focus on RelField [29], Open3DSG [28], and ConGraph [15], all of which leverage VLM-LLM pipelines with publicly available implementations. We do not include OpenFunGraph [69] (CVPR ’25), as it targets *functional* scene graphs with data and task definitions that are incompatible with 3DSSG.

Implementation Details. All models are implemented in PyTorch and trained on four NVIDIA RTX 4090 GPUs. We follow the standard OV protocol, holding out one-third of predicate classes as unseen and masking them during training. Unless stated otherwise, we optimize with Adam (learning rate 1×10^{-3} , batch size 32), train for 100–200 epochs, and select the best checkpoint by validation performance. The loss trade-off is $\lambda_{\text{geo}} : \lambda_{\text{rel}} = 0.4 : 0.6$, and the diffusion process uses $T = 150$ steps. The G^2T module uses width 512 with 8 attention heads; the Mixture-of-Experts has 5 experts. For discretization, geometry is vector-quantized with three independent codebooks (size 128 each), yielding six tokens per node (two each for position/size/angle). Relation semantics are quantized with a single codebook (size 64) and two tokens.

5.2. Comparison with the State-of-the-art

Comparisons with closed and OV 3DSSG methods. Tab. 1 reports predicate and relationship results under both closed-set and OV protocols. In the OV regime, GEODE establishes new state of the art on all metrics. Compared to the strongest open baseline, RelField [29], GEODE improves predicate R@3/R@5 by about 7 points and mR@3/mR@5 by roughly 6 points, and also yields clear gains on relationship prediction, boosting R@50/R@100 by 3/6 points

Table 1. Predicate and relationship prediction performance of GEODE and state-of-the-art 3DSGG methods under closed-set and open-set settings. [†] indicates results reproduced from released code, and the second-best result is underlined.

Models	Predicate				Relationship			
	Closed-Set		Open-Set		Closed-Set		Open-Set	
	R/mR@3	R/mR@5	R/mR@3	R/mR@5	R/mR@50	R/mR@100	R/mR@50	R/mR@100
Closed 3DSGG Methods								
3DSSG [57] (CVPR’2020)	0.89 / 0.54	<u>0.93</u> / 0.56	0.03 / 0.01	0.04 / 0.02	0.40 / 0.26	0.66 / 0.42	0.02 / 0.01	0.03 / 0.01
SGFN [59] (CVPR’2021)	0.97 / 0.68	0.99 / 0.69	0.05 / 0.02	0.07 / 0.03	0.85 / 0.51	0.87 / 0.55	0.04 / 0.02	0.05 / 0.03
SGRec3D [27] (WACV’2024)	0.96 / 0.58	0.99 / 0.59	0.07 / 0.03	0.09 / 0.05	0.89 / 0.54	0.91 / 0.56	0.05 / 0.03	0.07 / 0.04
VL-SAT [58] (CVPR’2023)	<u>0.98</u> / 0.60	0.99 / 0.62	0.10 / 0.05	0.12 / 0.06	0.90 / 0.56	0.93 / 0.57	0.07 / 0.05	0.10 / 0.06
SGFormer [36] (AAAI’2024)	<u>0.98</u> / 0.61	0.99 / 0.63	0.28 / 0.16	0.30 / 0.17	0.91 / 0.57	0.93 / 0.60	0.25 / 0.12	0.26 / 0.15
3D-HetSGP [37] (ECCV’2024)	<u>0.98</u> / 0.62	0.99 / 0.63	0.12 / 0.06	0.13 / 0.08	0.91 / 0.58	0.94 / 0.60	0.09 / 0.05	0.10 / 0.06
SCRSSG [65] (ICCV’2025)	<u>0.98</u> / 0.64	0.99 / 0.65	0.14 / 0.08	0.16 / 0.10	0.93 / 0.60	<u>0.95</u> / 0.62	0.12 / 0.07	0.13 / 0.08
OCRL [23] (NeurIPS’2025)	0.99 / <u>0.72</u>	0.99 / <u>0.73</u>	0.15 / 0.09	0.16 / 0.11	<u>0.95</u> / <u>0.68</u>	<u>0.95</u> / <u>0.69</u>	0.11 / 0.07	0.13 / 0.09
Open 3DSGG Methods								
CCL-3DSGG [5] (CVPR’2024)	0.44 / 0.30	0.45 / 0.31	0.43 / 0.28	0.45 / 0.29	0.41 / 0.27	0.43 / 0.29	0.38 / 0.26	0.40 / 0.27
ConceptGraphs [†] [15] (ICRA’2024)	0.48 / 0.35	0.49 / 0.35	0.47 / 0.32	0.48 / 0.34	0.41 / 0.31	0.43 / 0.32	0.40 / 0.30	0.42 / 0.32
Point2Graph [62] (CVPR’2025)	0.50 / 0.36	0.52 / 0.39	0.49 / 0.36	0.51 / 0.38	0.45 / 0.33	0.46 / 0.35	0.42 / 0.32	0.43 / 0.32
Open3DSG [†] [28] (CVPR’2024)	0.60 / 0.45	0.66 / 0.52	0.59 / 0.46	0.64 / 0.50	0.54 / 0.41	0.55 / 0.43	0.50 / 0.40	0.54 / 0.42
OW3DSGG [67] (AAAI’2026)	0.63 / 0.45	0.67 / 0.46	0.62 / 0.44	0.66 / 0.45	0.56 / 0.37	0.58 / 0.39	0.52 / 0.36	0.55 / 0.38
RelField [†] [29] (CVPR’2025)	0.74 / 0.60	0.83 / 0.68	<u>0.72</u> / <u>0.59</u>	<u>0.76</u> / <u>0.63</u>	0.74 / 0.61	0.75 / 0.62	<u>0.73</u> / <u>0.60</u>	<u>0.74</u> / <u>0.62</u>
GEODE	0.99 / 0.75	0.99 / 0.76	0.79 / 0.65	0.83 / 0.69	0.96 / 0.73	0.97 / 0.73	0.76 / 0.65	0.80 / 0.66

Table 2. Comparing spatial predicates with various SOTA 3DSGG models under the open-vocabulary setting.

Methods	R/mR@1	R/mR@5	R/mR@10
Open3DSG [28]	0.06 / 0.04	0.19 / 0.15	0.26 / 0.21
ConGraph [15]	0.10 / 0.08	0.24 / 0.19	0.33 / 0.25
RelField [29]	0.12 / 0.09	0.22 / 0.17	0.30 / 0.24
GEODE	0.31 / 0.26	0.87 / 0.71	0.90 / 0.75

and mR@50/mR@100 by about 4–5 points. These improvements on the more stringent triplet metrics indicate that GEODE produces more reliable geometry-consistent relations, especially for zero-shot pairs. In the conventional closed-set setting, GEODE matches or surpasses the best fully-supervised 3DSGG model (OCRL) on predicate recall and achieves noticeably higher mean-recall on relationships. Taken together, the results show that GEODE offers a unified improvement in both closed-set and OV 3DSGG, suggesting that its generative, geometry-first modeling leads to a stronger 3D scene-graph prior rather than overfitting to a particular vocabulary.

Spatial predicate result. Under the OV spatial-predicate benchmark (Tab. 2), GEODE clearly outperforms all VLM-LLM-based baselines on both recall and mean recall. Compared to RelField, R/mR@1 improves from 0.12/0.09 to 0.31/0.26, and R/mR@10 from 0.30/0.24 to 0.90/0.75, with similar margins over ConGraph and Open3DSG. These gains indicate that GEODE not only

recovers more correct spatial predicates higher but also recovers a broader set of rare predicates. We attribute this to GEODE’s geometry modeling: instead of encoding coordinates as soft textual hints, GEODE tokenizes 3D geometry and applies an intra-step geometry-to-relation conditioning scheme.

5.3. Ablation Study

We ablate the key inputs of GEODE: scene condition $\mathbf{x}$ (replacing text with image features), node feature (add node visual feature into the generation), geometric tokens Z_G (removed it at generation), as well as the within-step denoising order (geometry $\rightarrow$ relation vs. relation $\rightarrow$ geometry), and report Predicate R@K and Spatial R@K in Tab. 3.

Visual features as generative tokens. To examine whether appearance cues help open-vocabulary predicate prediction, we introduce a w/vision variant that tokenizes object visual embeddings from the OpenSeg[14] into node-level codes and jointly denoises these visual tokens together with geometric codes. The visual embeddings are discretized using the same quantization scheme as predicate tokens. As summarized in Tab. 3, adding visual tokens yields only marginal gains on overall predicate recall (R@3/R@5: 0.80/0.85 vs. 0.79/0.83) and offers no real benefit for spatial predicates (R@1/R@10: 0.27/0.88 vs. 0.31/0.90), while introducing a substantial increase in inference cost. These results suggest that explicit geometry already provides the dominant signal for spatial reasoning, so we keep GEODE geometry-first and refrain from treating visual features as generative tokens.

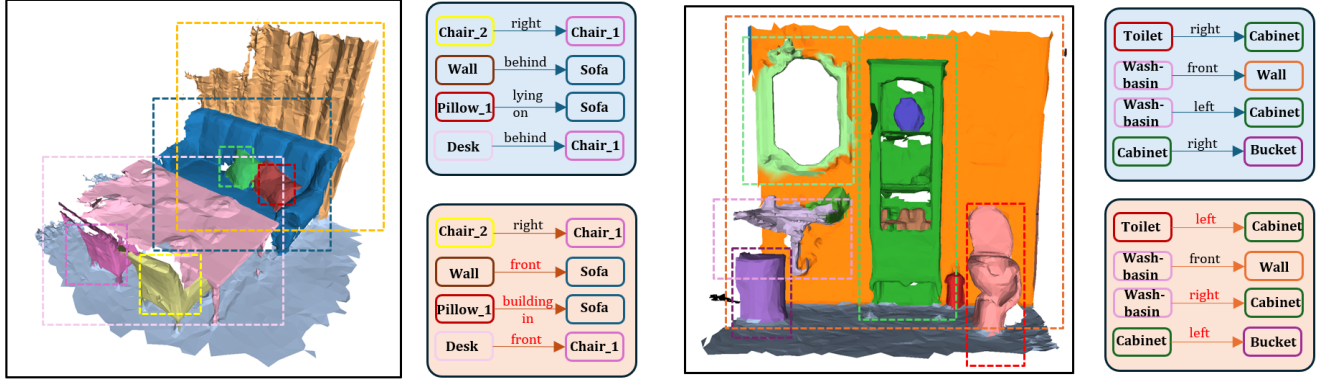


Figure 3. Qualitative open-vocabulary 3D scene graph predictions on 3DSSG [57]. For each scene, we show top-1 relationship predictions from RelField (bottom) and GEODE (top). Black predicates indicate correct relations; red predicates indicate errors.

Table 3. Ablation studies on the key components of GEODE.

Variants	Predicate		Spatial Predicate	
	R@3	R@5	R@1	R@10
w/vision	0.80	0.85	0.27	0.88
w/image	0.72	0.74	0.25	0.86
w/o Geo.	0.60	0.61	0.12	0.26
Pred. $\rightarrow$ Geo.	0.58	0.59	0.13	0.30
GEODE	0.79	0.83	0.31	0.90

Image features as Conditions. We further replace the textual scene title with CLIP features extracted from multi-view renderings (w/image). As shown in Tab. 3, this variant consistently underperforms text conditioning: predicate R@3/R@5 drop from 0.79/0.83 to 0.72/0.74, and spatial R@1/R@10 from 0.31/0.90 to 0.25/0.86. This suggests that global image embeddings provide noisier, less task-aligned signals, whereas the LLM-generated title offers a compact, interaction-focused description that better guides relation denoising.

Role of geometry. When we remove the geometry tokens Z_G and condition the diffusion process only on x and V (w/o Geo.), both predicate and spatial metrics degrade substantially (Tab. 3), with average drops of roughly 0.3 points. In this setting, the model falls back to global condition cues and label co-occurrence statistics, and struggles to produce reliable directional predicates. This confirms that discretized geometry, together with our geometry-first conditioning scheme, is essential for robust spatial reasoning.

Within-step update order. GEODE by default follows a geometry $\rightarrow$ relation schedule, first denoising geometry and then sampling predicates conditioned on the updated 3D layout (Eq. (9)). When we reverse this order (Pred. $\rightarrow$ Geo. in Tab. 3), both predicate and spatial

scores degrade noticeably, showing that predicting relations from a noisy state leads the model to rely on semantic shortcuts and then forces geometry to adapt to potentially inconsistent predicates. These results highlight that grounding relation updates on freshly recovered geometry is crucial for stable and accurate spatial reasoning.

5.4. Visualization Results

Figs. 3(a,b) present qualitative comparisons with RelField[29] on two challenging 3DSSG scenarios: near-depth ambiguity and symmetric layouts. In Fig. 3(a), pairs such as desktop-chairback and sofa-wall panel lie at very similar depths, under which RelField frequently confuses *in-front-of/behind* and *left/right*, whereas GEODE remains aligned with the true 3D ordering. In Fig. 3(b), a nearly symmetric configuration around a central wardrobe leads RelField to flip *left/right* labels, while GEODE’s orientation-aware modulation preserves consistent directional relations under symmetry.

6. Conclusion

We introduced **GEODE**, a geometry-guided discrete diffusion framework for open-vocabulary 3D scene graph generation. By vector-quantizing per-object 3D attributes and predicate semantics and imposing an intra-step geometry to relation denoising schedule, GEODE tightly couples spatial structure with relational prediction. The proposed G²T denoiser further integrates geometry-aware attention and MoE-based relation reasoning. On 3DSSG, GEODE achieves state-of-the-art performance in both closed-set and open-vocabulary settings, with especially pronounced improvements on spatial predicates and long-tailed relations.

Limitations and future work. Currently, geometry codebooks, diffusion schedules, and upstream detectors are optimized in separate stages. Future work will explore their end-to-end joint learning.

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